

δ -rings and λ -rings

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Abstract

We give the theory of Witt vectors a new formulation using the methods of universal algebra and category theory. We introduce an algebraic structure, that of a δ -ring. A δ -ring is a commutative ring equipped with an additional unary operation satisfying simple algebraic identities. The main result consists in showing that the forgetful functor from the category of δ -rings to commutative rings has a right adjoint W . An explicit computation allows the identification of $W(A)$ with the ring of Witt vectors over A . We give more conceptual proofs of several results of the theory of Witt vectors [7, 2].

1 δ -rings

Fix once and for all a prime number p and a power $q = p^r$ ($r \geq 0$).

Definition 1. A δ -ring A is a commutative ring equipped with a unary operation $\delta : A \rightarrow A$ satisfying the following identities:

1.

$$\delta(x + y) = \delta(x) + \delta(y) - \sum_{i=1}^{q-1} \frac{1}{p} \binom{q}{i} x^i y^{q-i};$$

2.

$$\delta(xy) = x^q \delta(y) + \delta(x) y^q + p \delta(x) \delta(y), \quad \delta(1) = 0.$$

We shall also say that A is a δ -ring with respect to the pair (p, q) . These conditions imply that the unary operation

$$f(x) = x^q + p \delta(x)$$

is a ring endomorphism of A :

$$f(x + y) = f(x) + f(y), \quad f(xy) = f(x)f(y), \quad f(1) = 1.$$

We say that f is the Frobenius endomorphism. Conversely, let f be an endomorphism of a commutative ring A without p -torsion. Suppose that for every $x \in A$,

$$f(x) \equiv x^q \pmod{pA}.$$

We obtain a δ -ring structure on A by defining δ as follows:

$$\delta(x) = \frac{1}{p}(f(x) - x^q).$$

Example 2. The ring $\mathbb{Z}$ admits a unique δ -ring structure since the only endomorphism of $\mathbb{Z}$ is the identity.

Example 3. Let L be a Galois extension of a field K . Let $B \subset L$ be a discrete valuation ring and let $A = B/\mathfrak{p}$. Suppose that p is a uniformizer for B and that $A/pA = \mathbb{F}_q$. One can show [1] the existence of a Frobenius substitution $\sigma \in \text{Gal}(L/K)$ characterised by

1. $\sigma(B) = B$;
2. $\sigma(x) \equiv x^q \pmod{\mathfrak{p}B}$.

A morphism of δ -rings is a ring homomorphism which preserves δ . We shall also call it a δ -morphism. We note that the Frobenius endomorphism is a δ -morphism.

Theorem 4. Let $\mathbb{Z}[x_0, x_1, \dots]$ be the polynomial ring on an infinite sequence of indeterminates. There is a unique δ -ring structure on $\mathbb{Z}[x_0, x_1, \dots]$ such that $\delta(x_n) = x_{n+1}$ for all n . Equipped with this structure, $\mathbb{Z}[x_0, x_1, \dots]$ is the free δ -ring on x_0 .

Let $\mathcal{A}$ denote the category of rings and let $\delta\mathcal{A}$ denote the category of δ -rings.

Theorem 5. The forgetful functor $U : \delta\mathcal{A} \rightarrow \mathcal{A}$ has a right adjoint $W : \mathcal{A} \rightarrow \delta\mathcal{A}$.

Category theory [6] brings to light the equivalence between the preceding theorem and the following one.

Theorem 6. Let $A \rightarrow B$ and $A \rightarrow C$ be morphisms of δ -rings. Consider the pushout square

$$\begin{array}{ccc} A & \rightarrow & C \\ \downarrow & & \downarrow \\ B & \rightarrow & B \otimes_A C \end{array}$$

There is a unique δ -ring structure on $B \otimes_A C$ such that the canonical morphisms $B \rightarrow B \otimes_A C$ and $C \rightarrow B \otimes_A C$ are δ -morphisms.

One can give a direct proof of these theorems or else use [8, 9]. As a consequence, we have:

Corollary 7. The functor $W : \mathcal{A} \rightarrow \delta\mathcal{A}$ is representable by the ring $\mathbb{Z}[x_0, x_1, \dots]$. There is a natural bijection

$$\gamma_A : W(A) \cong A^{\mathbb{N}}.$$

The bijection γ_A is made explicit as follows:

$$W(A) = \text{Hom}_{\delta\mathcal{A}}(\mathbb{Z}[x_0, x_1, \dots], W(A)) \cong \text{Hom}_{\mathcal{A}}(\mathbb{Z}[x_0, x_1, \dots], A) \cong A^{\mathbb{N}}.$$

2 Witt vectors

To see that $W(A)$ is isomorphic to the ring of Witt vectors over A , it suffices to use a different description of the free δ -ring on one generator. This description is based on the following result.

Proposition 8. *In the theory of δ -rings, one can define a unique sequence of unary operations $\delta_0, \delta_1, \delta_2, \dots$ satisfying the identities*

$$f^n(x) = \delta_0(x)^{q^n} + p\delta_1(x)^{q^{n-1}} + \dots + p^n\delta_n(x) \quad (n \geq 0).$$

The first two terms of the sequence are $\delta_0(x) = x$ and $\delta_1(x) = \delta(x)$.

Proposition 9. *There is a unique δ -ring structure on $\mathbb{Z}[Y_0, Y_1, Y_2, \dots]$ such that $\delta_n(Y_0) = Y_n$ for all $n \geq 0$. Equipped with this structure, $\mathbb{Z}[Y_0, Y_1, Y_2, \dots]$ is the free δ -ring on Y_0 .*

Theorem 10. *If $q = p$, the ring $W(A)$ is identified with the ring of Witt vectors over A .*

To each element $x \in W(A)$ corresponds a Witt vector $\pi_A(x) \in A^{\mathbb{N}}$. The description of π_A is analogous to that of γ_A but uses the free δ -ring $\mathbb{Z}[Y_0, Y_1, \dots]$ instead.

The composite endofunctor $\mathcal{A} \rightarrow \delta\mathcal{A} \rightarrow \mathcal{A}$ is a comonad [6, 3, 2] on $\mathcal{A}$. For simplicity, we still denote this functor by W . There are natural transformations

$$W(A) \xrightarrow{\varepsilon_A} A, \quad W(A) \xrightarrow{\Delta_A} W(W(A))$$

satisfying associativity and unity conditions. The homomorphism Δ is closely related to the Artin–Hasse exponential [3]. A δ -ring structure on A is equivalent to a coalgebra structure [6] $\alpha : A \rightarrow W(A)$. This is described as follows: for every $x \in A$, we have

$$\pi_A(\alpha(x)) = (\delta_0(x), \delta_1(x), \dots).$$

Remark 11. *Let A be a ring without p -torsion equipped with an endomorphism f such that for every $x \in A$, $f(x) \equiv x^p \pmod{pA}$. The Dieudonné–Cartier lemma [4, page 508] asserts the existence of a homomorphism $s : A \rightarrow W(A)$. In the present context, this result is interpreted as follows: the hypotheses imply the existence of a δ -ring structure on A , and one verifies that $s = \alpha$.*

Remark 12. *Under the hypotheses of the preceding remark, we give a particularly simple description of $W(A)$ [5]. Set*

$$B = \{(a_n) \in A^{\mathbb{N}} \mid a_{n+1} \equiv f(a_n) \pmod{p^{n+1}} \forall n\}.$$

One verifies that B is a subring of the product ring $A^{\mathbb{N}}$. The translation operator $t((a_n)) = (a_{n+1})$ is an endomorphism of B , and for every $x \in B$,

$$t(x) \equiv x^p \pmod{pB}.$$

Since B is p -torsion-free, we obtain a δ -ring structure on B for which t is the Frobenius endomorphism. One then checks directly that with the projection $B \rightarrow A$, the ring B becomes the cofree δ -ring over the ring A , hence $B \cong W(A)$.

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